SOCIOL 723: Social Statistics II

Week 2, Regression Inference and Robust Standard Errors

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Today

The Sampling Distribution of OLS

Hypothesis Testing

Heteroskedasticity and Robust Standard Errors

Clustered Data

The Bootstrap

What Robust Standard Errors Cannot Fix

★ = essential, you must understand this ★ = for understanding, not examined

Where We Left Off

Last week, for the bivariate model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \bar{X})\,\epsilon_i}{\sum_i (X_i - \bar{X})^2}, \qquad \mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sum_i (X_i - \bar{X})^2\ \mathrm{Var}(\epsilon_i \mid X)}{\left[\sum_i (X_i - \bar{X})^2\right]^2},$$

which collapses to $\sigma^2 / \sum_i (X_i - \bar{X})^2$ only if $\mathrm{Var}(\epsilon_i \mid X)$ is the same for every i.

Today's agenda

1. What is the sampling distribution of $\hat{\beta}_1$, with and without normality?
2. How do we test hypotheses, single, joint, and nonlinear?
3. What if $\mathrm{Var}(\epsilon_i \mid X)$ is *not* constant? (Heteroskedasticity, then clustering.)
4. The bootstrap as a general-purpose alternative.
5. What robust standard errors can and cannot fix.

Reminder: the Two-Star Convention

Frames marked ★ are **essential**: problem sets and exams draw on them directly. Frames marked ★ are here for *understanding*, not for evaluation: matrix derivations, asymptotic arguments, and the general formulas you will meet in the literature.

- Examinable this week: the scalar variance formula and where it comes from; what heteroskedasticity and clustering *are* and why they matter; how to read and choose a standard error; what a bootstrap does; and the list of problems robust standard errors do not solve.
- Every "robust" standard error in this course is the *same* scalar formula with $\text{Var}(\epsilon_i \mid X)$ estimated differently. If you hold on to that one sentence, none of this is complicated.

Sampling Distribution

The Question, in Plain Words ★

Last week we computed $\hat{\beta}_1 = 0.119$: one more year of schooling goes with about 12% higher earnings in our sample of 3,509 people.

- The GSS could just as easily have interviewed a *different* 3,509 people. That sample would have produced a different number, perhaps 0.112, perhaps 0.127.
- So 0.119 is one draw from a lottery. Before treating it as evidence about the population, we need to know how much it moves from draw to draw.
- The standard error is exactly that: the give-or-take attached to the estimate. Formally, it is (an estimate of) the standard deviation of $\hat{\beta}_1$ across hypothetical repeated samples.

This week in one sentence

Everything called "statistical inference", standard errors, t-statistics, p-values, confidence intervals, is machinery for learning that give-or-take number from the single sample we actually have.

What a Sampling Distribution Is ★

$\hat{\beta}_1$ is a number computed from *your* sample. Had you drawn a different sample from the same population, you would have got a different number.

The thought experiment

Imagine drawing sample after sample from the population and computing $\hat{\beta}_1$ each time. The histogram of those numbers is the sampling distribution. Where it is centred tells you about bias; how wide it is *is* the standard error.

- You only ever see one draw. Everything in this lecture is about inferring the shape of that histogram from a single sample.
- In lab we run the experiment by simulation, which is the fastest way to make the idea concrete.

Two Theorems We Lean On All Semester ★

Both are statements about what happens to a *sample average* as n grows.

Law of large numbers (LLN)

The average of n independent draws settles down on the population mean: $\frac{1}{n}\sum_i Z_i \xrightarrow{p} E[Z]$. Flip a fair coin 10 times and the share of heads could be anything; flip it 10,000 times and it sits within a whisker of 0.5.

Central limit theorem (CLT)

Not only does the average settle down; the *shape* of its wobble around the truth becomes a bell curve, $\sqrt{n}\,(\bar{Z} - E[Z]) \xrightarrow{d} \mathcal{N}(0, \mathrm{Var}(Z))$, whatever the distribution of Z itself, skewed, lumpy, or binary, so long as its variance is finite.

Why we care: $\hat{\beta}_1$ is built out of sample averages, so the LLN will give consistency and the CLT approximate normality. We verify both by simulation in Thursday's lab; watching the bell curve emerge once is worth a dozen proofs.

Route 1: Exact, Under Normality

Assume A1–A6, including $\epsilon_i \mid X \sim \mathcal{N}(0, \sigma^2)$. Since $\hat{\beta}_1 - \beta_1$ is a *weighted sum of the errors*, it is normal:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \ \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}\right), \qquad t = \frac{\hat{\beta}_1 - \beta_1}{\mathrm{se}(\hat{\beta}_1)} \sim t_{n-k}.$$

- The t (rather than the normal) appears because we replaced the unknown σ^2 by the estimate s^2.
- Exact at any n, but it requires normal errors, which is rarely credible for earnings, counts, durations, or attitude scales.

Route 2: Approximate, Without Normality ★

Drop A6 and look again at

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n}\sum_i (X_i - \bar{X})\epsilon_i}{\frac{1}{n}\sum_i (X_i - \bar{X})^2}.$$

- The **denominator** is a sample average. By the law of large numbers it settles down to $\text{Var}(X_i)$ as n grows.
- The **numerator** is also a sample average, of the mean-zero terms $(X_i - \bar{X})\epsilon_i$. By the central limit theorem, sample averages of i.i.d. mean-zero terms are approximately normal once n is large, *whatever* the distribution of ϵ_i.

So $\hat{\beta}_1$ is approximately normal in large samples with no assumption about the shape of the error distribution. This is the modern default, and it is the route that generalizes to MLE (Week 3), IV, and everything after.

Testing

Single Restrictions ★

To test $H_0 : \beta_j = \beta_j^0$,

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-k} \text{ (exactly, under A6) or } \mathcal{N}(0,1) \text{ (approximately, for large } n).$$

A 95% confidence interval is $\hat{\beta}_j \pm 1.96\,\mathrm{se}(\hat{\beta}_j)$.

- The t_{n-k} and $\mathcal{N}(0,1)$ critical values differ by less than 2% once $n-k > 100$. Reporting t is a convention, not a claim to exactness.
- Everything depends on whether se is right. A t of 3.0 computed from a standard error that is half of what it should be is really a t of 1.5.

Joint Restrictions: the F-Test *Is* Model Comparison ★

To test q restrictions at once (“neither `wordsum` nor `pareduc` matters”), fit the restricted (compact) and unrestricted (augmented) models and compare their errors:

$$F = \frac{(SSE_C - SSE_A)/q}{SSE_A/(n-k)} \sim F_{q,\,n-k} \qquad \text{under A1–A6.}$$

Exactly the model-comparison F from *Social Statistics I*.

An important caveat

The sums-of-squares form is valid only under homoskedasticity. With robust or clustered standard errors you must use the Wald form, `car::linearHypothesis(..., vcov = )` or `lmtest::waldtest(..., vcov = )`. Plain `anova()` on two `lm` objects gives you the *non-robust* test. This is a very common error in published work.

The Wald Statistic ★

Write q linear restrictions as $\boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{q}$ with $\boldsymbol{R}$ a $q \times k$ matrix of rank q:

Hypothesis	$\boldsymbol{R}$, $\boldsymbol{q}$
$\beta_2 = 0$	$\boldsymbol{R} = (0, 1, 0, \dots)$, $\boldsymbol{q} = 0$
$\beta_2 = \beta_3$	$\boldsymbol{R} = (0, 1, -1, 0 \dots)$, $\boldsymbol{q} = 0$
$\beta_2 = \beta_3 = 0$	two rows, $\boldsymbol{q} = (0, 0)'$
All slopes zero	$\boldsymbol{R} = [\boldsymbol{0} \;\; \boldsymbol{I}_{k-1}]$

$$W = (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{q})' \left[\boldsymbol{R}\,\hat{V}(\hat{\boldsymbol{\beta}})\,\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{q}) \xrightarrow{d} \chi^2_q, \qquad F = W/q.$$

Substituting any $\hat{V}(\hat{\boldsymbol{\beta}})$ you like (robust, clustered) gives the corresponding robust joint test.

Nonlinear Restrictions: the Delta Method ★

For a smooth function $g(\boldsymbol{\beta})$, a ratio of coefficients, a turning point $-\beta_1/2\beta_2$, a marginal effect:

$$\hat{V}(g(\hat{\boldsymbol{\beta}})) = \nabla g(\hat{\boldsymbol{\beta}})' \, \hat{V}(\hat{\boldsymbol{\beta}}) \, \nabla g(\hat{\boldsymbol{\beta}}).$$

- Intuition: a smooth function is locally a straight line, and we know the variance of a linear function. So linearize, then use what we know.
- Implemented as `car::deltaMethod()` or the `marginaleffects` package.
- First-order approximation: it can be poor when g is strongly nonlinear near $\hat{\boldsymbol{\beta}}$, or when the denominator of a ratio is close to zero. The bootstrap is often better there.
- We use it constantly in Week 4 for marginal effects and predicted probabilities.

A Note on Multiple Testing

Everything so far controlled the error rate for *one* planned comparison. The moment you look at many outcomes, many subgroups, or many specifications, the guarantee has to hold across the whole family of claims, and it does not.

If you run m independent tests at level α, the chance of at least one false rejection is $1 - (1 - \alpha)^m$: for $m = 20$, about 64%.

- **Bonferroni**: test each at α/m. Controls the family-wise error rate; conservative.
- **Holm**: sequentially rejective; uniformly more powerful than Bonferroni with the same guarantee.
- **Benjamini–Hochberg**: controls the *false discovery rate*, the right tool with many outcomes or subgroups when you can tolerate some false positives.
- `p.adjust(p, method = "holm" / "BH")`.

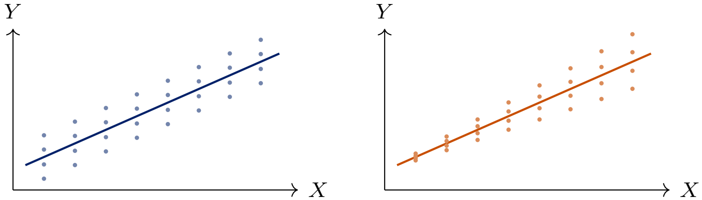
A Note on Multiple Testing (cont.)

Sociologically relevant cases: many outcomes, many subgroups, many specifications. Pre-registering a small set of primary hypotheses is a better solution than any correction.

Heteroskedasticity

What Heteroskedasticity Is ★

Homoskedasticity (A5) says $\mathrm{Var}(\epsilon_i \mid X_i) = \sigma^2$: the spread of the outcome around the line is the same everywhere.



homoskedastic heteroskedastic

Under exogeneity, both lines are estimated without bias: heteroskedasticity by itself does not bias $\hat{\beta}$. Only the *uncertainty* about them differs, and only the right-hand panel needs a different formula.

Heteroskedasticity Is the Norm

- Earnings dispersion rises steeply with education and experience: at 8 years of schooling almost everyone earns a modest wage; at 20 years the range is enormous.
- Group-level outcomes built from different numbers of individuals, a county mean over 50 people is noisier than one over 50,000.
- Binary outcomes: $\mathrm{Var}(Y_i \mid X_i) = p(X_i)(1 - p(X_i))$ is *necessarily* heteroskedastic, so the linear probability model always violates A5.

What it does and does not do

Does not: bias $\hat{\beta}$, or break consistency.
Does: invalidate the standard errors your software prints by default, so t-statistics, p-values and confidence intervals are wrong; and cost efficiency (OLS is no longer BLUE).

The Scalar Sandwich ★

Go back to the exact variance, *without* assuming constant variance. Write $\sigma_i^2 \equiv \mathrm{Var}(\epsilon_i \mid X)$ and $d_i \equiv X_i - \bar{X}$:

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \mathrm{Var}\left(\frac{\sum_i d_i \epsilon_i}{\sum_i d_i^2} \;\middle|\; X \right) = \frac{\sum_i d_i^2\, \sigma_i^2}{\left(\sum_i d_i^2\right)^2}.$$

- If $\sigma_i^2 = \sigma^2$ for all i, it factors out and we recover $\sigma^2 / \sum_i d_i^2$.
- If not, the numerator is a d_i^2-weighted average of the individual error variances. Observations far from $\bar{X}$, the high-leverage ones, get the most weight.
- This is the entire idea. The “bread” is $\sum_i d_i^2$, twice; the “meat” is $\sum_i d_i^2 \sigma_i^2$.

White's Estimator, in Scalar Form ★

We cannot observe σ_i^2. But we have $\hat{e}_i^2$, and $E[\hat{e}_i^2] \approx \sigma_i^2$. Plug it in:

$$\boxed{\hat{V}_{\text{HC0}}(\hat{\beta}_1) = \frac{\sum_i d_i^2\, \hat{e}_i^2}{\left(\sum_i d_i^2\right)^2}} \qquad \text{versus} \qquad \hat{V}_{\text{classical}}(\hat{\beta}_1) = \frac{s^2}{\sum_i d_i^2}.$$

- That is the whole of "heteroskedasticity-robust," "Huber–White," "Eicker–Huber–White," and the `robust` option in `Stata`.
- Due to White (1980), building on Eicker (1967) and Huber (1967).
- Notice the difference: the classical formula uses *one* pooled s^2; the robust formula lets every observation carry its own $\hat{e}_i^2$.

"But You Only Have One Residual Per Observation"

A frequent and reasonable objection: how can we estimate n different variances from n observations?

- We never estimate σ_i^2 consistently for any particular i, and we do not need to.
- The object we need is the *sum* $\sum_i d_i^2 \sigma_i^2$: a single number. Each $\hat{e}_i^2$ is a noisy but roughly unbiased estimate of σ_i^2, and adding up n of them averages the noise away.
- Same logic that lets you estimate $\text{Var}(\bar{Y})$ from one sample: you never learn any individual's deviation, only the average square.

The general version ★: $\hat{\boldsymbol{\Omega}}^* = \mathbb{E}_n[\hat{e}_i^2 X_i X_i'] \xrightarrow{p} E[\epsilon_i^2 X_i X_i']$, so $\hat{V}_{\text{HC0}}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\left(\sum_i \hat{e}_i^2 X_i X_i'\right)(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

The General Statement ★

For the full coefficient vector, writing sample averages as $\mathbb{E}_n$:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \left(\mathbb{E}_n[X_i X_i']\right)^{-1} \mathbb{E}_n[X_i \epsilon_i].$$

The LLN gives $\mathbb{E}_n[X_i X_i'] \xrightarrow{p} E[X_i X_i'] \equiv \boldsymbol{Q}$ and $\mathbb{E}_n[X_i \epsilon_i] \xrightarrow{p} \mathbf{0}$, so $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}$ (*consistency*). Multiplying by $\sqrt{n}$ and applying the CLT to the second factor,

$$\boxed{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(\mathbf{0},\ \boldsymbol{Q}^{-1} \boldsymbol{\Omega}^* \boldsymbol{Q}^{-1}\right)}, \qquad \boldsymbol{\Omega}^* = E[\epsilon_i^2 X_i X_i'].$$

Note what is *not* assumed: no normality, no homoskedasticity. Note also the weaker exogeneity requirement, $E[X_i \epsilon_i] = 0$ suffices for consistency; $E[\epsilon_i \mid X_i] = 0$ is not needed.

Reading the "Sandwich" ★

$$\boldsymbol{V} = \underbrace{\boldsymbol{Q}^{-1}}_{\text{bread}} \underbrace{\boldsymbol{\Omega}^{*}}_{\text{meat}} \underbrace{\boldsymbol{Q}^{-1}}_{\text{bread}}, \qquad \boldsymbol{Q} = E[X_i X_i'], \quad \boldsymbol{\Omega}^{*} = E[\epsilon_i^2 X_i X_i'].$$

- The **bread** depends only on the distribution of the regressors.
- The **meat** is where every assumption about the error structure lives. Classical, robust, clustered, and HAC standard errors differ only in how the meat is estimated.
- If $E[\epsilon_i^2 \mid X_i] = \sigma^2$ then $\boldsymbol{\Omega}^{*} = \sigma^2 \boldsymbol{Q}$ and $\boldsymbol{V} = \sigma^2 \boldsymbol{Q}^{-1}$, the classical formula.

The organizing idea for the whole lecture

Keep the estimator $\hat{\beta}$; change how you estimate the meat.

Finite-Sample Corrections: HC0–HC3 ★

$\hat{e}_i^2$ is biased downward: *under homoskedasticity* $E[\hat{e}_i^2] = \sigma^2(1 - h_{ii})$, where $h_{ii} = X_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}X_i$ is observation i's leverage (large when its covariates are unusual). So HC0 *understates* standard errors in small samples. The fixes rescale the squared residuals (motivated by the homoskedastic benchmark, used regardless):

	Meat uses	Comment
HC0	$\hat{e}_i^2$	White's original; anticonservative when n is small
HC1	$\frac{n}{n-k}\hat{e}_i^2$	simple d.f. correction; `Stata`'s `robust` default
HC2	$\dfrac{\hat{e}_i^2}{1-h_{ii}}$	unbiased under homoskedasticity
HC3	$\dfrac{\hat{e}_i^2}{(1-h_{ii})^2}$	approximates the jackknife; recommended default

MacKinnon and White (1985) and Long and Ervin (2000): with $n < 250$ or high leverage, use **HC3**. All four coincide as $n \to \infty$.

How Much Does It Matter?

- Robust standard errors can be *larger or smaller* than classical ones. There is no general direction.
- The difference is driven by whether the error variance happens to be large where X is extreme. In our GSS earnings regression it is modest: $\text{se}(\hat{\beta}_{\texttt{educ}})$ goes from 0.00510 (classical) to 0.00547 (HC0) to 0.00549 (HC3), about 7%.
- With a skewed or sparse regressor, a rare binary treatment, a long-tailed count; the difference can be a factor of two.
- Clustering, by contrast, routinely changes standard errors by factors of three to ten. In most sociological data that is the bigger issue.

Should You *Test* for Heteroskedasticity?

Classical tests exist: Breusch–Pagan, White.

Recommendation: no

Do not pre-test and then choose. Just report robust standard errors.

- A pre-test makes your final inference conditional on the pre-test outcome, which distorts the very sampling distribution you are claiming.
- The cost of robust standard errors when the errors happen to be homoskedastic is small, a modest efficiency loss in the *variance estimate*.
- The cost of classical standard errors when they are wrong is invalid inference.
- Exception: the tests are useful *diagnostically*. Heteroskedasticity is often a symptom of a misspecified functional form, which is worth knowing about for its own sake.

Weighted Least Squares: the Road Not Taken

If we knew $\text{Var}(\epsilon_i \mid X_i) = \sigma^2/w_i$, weighted least squares would restore efficiency: minimize $\sum_i w_i(Y_i - X_i'b)^2$.

- BLUE under the *correct* weights, Gauss–Markov applied to the transformed model.
- But we almost never know the weights, and estimating them ("feasible GLS") introduces problems of its own.
- Worse: if the CEF is not exactly linear, OLS and WLS estimate *different* weighted approximations to it. They answer different questions, so a discrepancy between them is not a bug to be fixed.
- Modern practice: keep OLS, fix the standard errors. Use weights when they come from the *design* (survey sampling weights, group sizes), not from a guess about the error variance.

Clustering

Why Independence Fails ★

Sociological data are almost never i.i.d. draws:

- Students within schools; workers within firms; siblings within families.
- Repeated observations on the same individual (panel data).
- Multi-stage survey designs, the GSS samples tracts, then households.
- Treatment assigned at the group level (a state law), outcomes measured at the individual level.

Error components model

$$Y_{ig} = X_{ig}'\beta + \underbrace{\nu_g}_{\text{cluster shock}} + \underbrace{\eta_{ig}}_{\text{idiosyncratic}}, \qquad \rho_e = \frac{\sigma_\nu^2}{\sigma_\nu^2 + \sigma_\eta^2}$$

Everyone in cluster g shares the shock ν_g, so their errors are correlated. ρ_e is the intra-class correlation.

Where the Scalar Formula Breaks ★

The variance of $\hat{\beta}_1$ involves *all* the covariances of the errors:

$$\operatorname{Var}\left(\sum_i d_i \epsilon_i\right) = \underbrace{\sum_i d_i^2\, \sigma_i^2}_{\text{what HC0 estimates}} + \underbrace{\sum_i \sum_{j \neq i} d_i d_j \operatorname{Cov}(\epsilon_i, \epsilon_j)}_{\text{assumed zero so far}}.$$

- Independent sampling makes the second term vanish. Clustering does not.
- If errors within a cluster are *positively* correlated and d_i has the same sign within clusters, every one of those cross terms is positive , so the true variance is larger than the formula we have been using, often by a lot.
- There are on the order of $n(\bar{n} - 1)$ such cross terms versus n terms in the first sum, so even tiny correlations can dominate.

The Moulton Factor: How Bad Is It? ★

With equal cluster sizes $\bar{n}$ and an error-components structure, Moulton (1986) showed

$$\frac{\mathrm{Var}_{\mathrm{clustered}}(\hat{\beta}_1)}{\mathrm{Var}_{\mathrm{OLS}}(\hat{\beta}_1)} = 1 + (\bar{n} - 1)\,\rho_x\,\rho_e,$$

where ρ_x is the intra-class correlation of the regressor.

- If the regressor is *constant within clusters*, treatment assigned at the cluster level, then $\rho_x = 1$ and the inflation is $1 + (\bar{n} - 1)\rho_e$. This is the worst case.
- 50 states, 1,000 individuals each, $\rho_e = 0.01$ (small!): $\sqrt{1 + 999(0.01)} \approx 3.3$. Standard errors are more than three times too small, so a reported t of 6 is really a t of 1.8.
- Moral: the effective sample size is closer to the number of *clusters* than the number of observations.

Cluster-Robust Variance (Liang–Zeger) ★

Let $g = 1, \dots, G$ index clusters, with $\boldsymbol{X}_g$ and $\hat{\boldsymbol{e}}_g$ the stacked regressors and residuals in cluster g:

$$\boxed{\hat{V}_{\mathrm{CR}}(\hat{\boldsymbol{\beta}}) = c\,(\boldsymbol{X}'\boldsymbol{X})^{-1}\Big(\sum_{g=1}^{G} \boldsymbol{X}_g'\hat{\boldsymbol{e}}_g\hat{\boldsymbol{e}}_g'\boldsymbol{X}_g\Big)(\boldsymbol{X}'\boldsymbol{X})^{-1}}$$

with the usual small-sample scaling $c = \frac{G}{G-1} \cdot \frac{n-1}{n-k}$ (CR1; `Stata`'s default).

- Same sandwich; the meat now sums outer products of *cluster* score vectors instead of individual ones, which is exactly how the cross terms get included.
- Allows arbitrary correlation within cluster (including serial correlation over time) and any heteroskedasticity, while assuming independence *across* clusters.
- Consistency is in $G \to \infty$, not $n \to \infty$.

At What Level Should You Cluster? ★

Abadie et al. (2023): clustering is a statement about the *design*, not a knob to tune.

- **Sampling-based**: cluster at the level of the sampling units, when the clusters in your data are a small share of the clusters in the population.
- **Design-based**: cluster at the level at which *treatment* is assigned. If a policy varies by state, cluster on state, even with individual-level data.
- Cluster at the *coarsest* level for which you have a plausible argument; never finer than the treatment assignment.
- "Cluster on everything" is not free: with few clusters the estimator is biased downward and the test over-rejects.
- Never choose the level by whichever gives the smallest p-value.

The Few-Clusters Problem

With small G, $\hat{V}_{\text{CR}}$ is biased downward and the reference distribution is too thin. Worry below $G \approx 40$–50; be alarmed below 20. The symptom is rejection rates of 10–15% for a nominal 5% test.

Remedies

1. **CR2 with Satterthwaite degrees of freedom** (Bell and McCaffrey 2002; Imbens and Kolesar 2016): a leverage correction (the cluster analogue of HC2) plus data-dependent d.f. `clubSandwich::coef_test()`.
2. **Wild cluster bootstrap-*t*** (Cameron et al. 2008): impose H_0, resample cluster-level ± 1 weights, and build the reference distribution empirically. `fwildclusterboot`.
3. **Aggregate**: collapse to G cluster-level observations and analyze those. Crude but honest.
4. **Randomization inference** when treatment is assigned to few units.

Serial Correlation: Bertrand, Duflo, and Mullainathan (2004)

A famous demonstration of what happens when you ignore within-unit correlation.

- They ran *placebo* difference-in-differences: take CPS data, invent a state-level "law" beginning in a random year, and estimate its effect on women's wages. The true effect is zero by construction.
- With conventional standard errors the fake law came out "significant" at 5% in 45% of the placebo runs.
- Cause: state-level outcomes are highly serially correlated over time, and the standard errors treated 50×21 observations as independent.
- Clustering by state fixed it, with 50 clusters, just barely enough.

We return to this in Week 10. Design and inference are not separable questions.

Other Dependence Structures ★

- **Two-way / multi-way clustering** (Cameron et al. 2011): firms *and* years, individuals *and* places. Add the two cluster-robust matrices and subtract their intersection.
- **HAC (Newey–West)**: time-series dependence of unknown form, handled with a bandwidth-weighted sum of autocovariances. For one long series rather than many short ones.
- **Spatial HAC (Conley)**: correlation decaying in geographic distance.
- **Network dependence**: correlation along a graph; an active area with fewer settled answers.

All are the same sandwich with a different meat. Once you can read $\hat{V}$, you can read any of these off the formula.

Bootstrap

The Idea

Everything so far derived a *formula* for the sampling distribution. The bootstrap instead simulates it.

Plug-in principle (Efron 1979)

We want the sampling distribution of a statistic when samples are drawn from the population F. We do not know F, but the sample *is* our best estimate of F. So draw new samples from the sample.

- In our notation: replace E by $\mathbb{E}_n$, and resample.
- Especially valuable for statistics with no convenient closed-form variance: quantiles, ratios, decompositions, two-step estimators, indirect effects.

The Nonparametric (Pairs) Bootstrap ★

1. For $b = 1, \ldots, B$: draw n observations (Y_i, X_i) *with replacement* from the sample.
2. Re-estimate on the resample to get $\hat{\theta}^{(b)}$.
3. Use the empirical distribution of $\{\hat{\theta}^{(b)}\}_{b=1}^{B}$.

- se_{boot} = the standard deviation of the $\hat{\theta}^{(b)}$.
- **Percentile CI**: the 2.5th and 97.5th percentiles of $\{\hat{\theta}^{(b)}\}$.
- Because it resamples (Y_i, X_i) jointly, the pairs bootstrap is automatically heteroskedasticity-robust, it makes no assumption about σ_i^2 at all.
- $B = 1{,}000$ for standard errors; $B \geq 5{,}000$ for percentile intervals or p-values.

Variants

Variant	What is resampled	When to use
Pairs	(Y_i, X_i) jointly	default; robust to heteroskedasticity
Residual	$\hat{e}_i$, then $Y_i^* = X_i'\hat{\boldsymbol{\beta}} + \hat{e}_i^*$	requires homoskedasticity; fixed design
Wild	$Y_i^* = X_i'\hat{\boldsymbol{\beta}} + \hat{e}_i v_i$, $v_i \in \{-1, 1\}$	heteroskedastic, small n
Block / cluster	whole clusters	clustered data
Wild cluster	cluster-level v_g, H_0 imposed	few clusters

Rademacher weights: $v_i = +1$ or -1 with probability $1/2$ each, so $E[v_i] = 0$ and $E[v_i^2] = 1$, the resampled errors keep the right variance observation by observation.

Confidence Intervals: Not All Equal

- **Normal-approximation**: $\hat{\theta} \pm 1.96\,\mathrm{se}_{\mathrm{boot}}$. Simple; assumes symmetry.
- **Percentile**: quantiles of $\{\hat{\theta}^{(b)}\}$. Handles skew and respects transformations.
- **BC$_a$**: bias-corrected and accelerated; second-order accurate and the usual recommendation for a percentile-type interval.
- **Bootstrap-t (studentized)**: bootstrap the t *statistic* $(\hat{\theta}^{(b)} - \hat{\theta})/\mathrm{se}^{(b)}$ and use its quantiles. **Asymptotic refinement**: error $O(n^{-1})$ instead of $O(n^{-1/2})$. Requires a standard error at each replication.

The wild cluster bootstrap-t is exactly this idea applied to the few-clusters problem, which is why it works so much better than a naive cluster bootstrap.

When the Bootstrap Fails

The bootstrap is not magic; it needs the statistic to be a *smooth* functional of the data.

- **Non-smooth estimators**: the sample maximum or minimum; nearest-neighbour matching estimators (Abadie and Imbens 2008).
- **Parameters on a boundary**, a variance component at zero.
- **Heavy tails** with infinite variance.
- **Dependence the resampling scheme does not preserve.** If you resample individuals when the data are clustered, the bootstrap reproduces the *wrong*, too-small standard error just as faithfully as the analytic formula does.

Bootstrap the design

Resample the units that were independently sampled. That single rule prevents most bootstrap mistakes.

Limits

Freedman's Critique

Freedman (2006) makes an uncomfortable point.

The argument

The sandwich gives a consistent estimate of the variance of $\hat{\beta}$ *around the population regression coefficient* β. If the model is misspecified, if the CEF is not $X_i'\beta$, then β is merely the best linear approximation, and a correct standard error for a quantity you do not care about is small comfort.

- In his words: "if the model is seriously in error, the sandwich may help on the variance side, but the parameters being estimated... are likely to be meaningless."
- If the model is right, you did not need the sandwich (much).
- If the model is wrong, the sandwich fixes the second-order problem (variance) and leaves the first-order problem (bias) untouched.

A Scoreboard ★

Problem	Robust SEs?	What actually helps
Heteroskedasticity	**yes**	nothing more needed
Within-cluster correlation	**yes** (cluster-robust)	design-based clustering
Serial correlation in panels	**yes**	cluster on unit
Non-normal errors	**yes** (asymptotically)	larger n
Omitted variable bias	no	identification strategy (Wks 5–11)
Simultaneity, reverse causation	no	IV, RDD, DID
Measurement error in X	no	IV, validation data
Wrong functional form	no	saturate, splines, flexible ML
Selection into the sample	no	modelling the selection
Wrong estimand	no	say what you are estimating
Influential outliers	no	diagnostics; robust regression
p-hacking, specification search	no	pre-registration

A Practical Protocol ★

1. State the estimand and the design *before* choosing a standard error.
2. **Cluster where treatment is assigned or where sampling occurred.** Otherwise use HC2 or HC3.
3. Report the number of clusters G. If $G < 40$, use CR2 with Satterthwaite d.f. or a wild cluster bootstrap, and say so.
4. Never pair a robust variance with a non-robust joint test (`anova()` defaults are not robust).
5. If robust and classical standard errors differ dramatically, treat it as a diagnostic: something about the specification or the influence structure deserves attention.
6. Report the specification you pre-committed to, then robustness checks, not the reverse.

Looking Ahead

- We now have a complete account of OLS: what it estimates, and how to quantify our uncertainty about it.
- **Weeks 3–4**: maximum likelihood. The same asymptotic machinery, score, information, sandwich, returns in a more general form, and OLS turns out to be a special case.
- **Weeks 5+**: the harder question. Not "how precise is $\hat{\beta}$?" but "is β the thing I want?"

Problem Set 1

Assigned today, due Monday September 14 at 11:59PM. Covers Weeks 1–2: scalar OLS algebra, FWL, omitted variable bias, and robust/clustered inference on the GSS extract.

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